Phasor Tip Speed Scaling at Fourier Jumps: A Phasor-Kinematic Interpretation of the Gibbs Phenomenon

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Abstract

The Gibbs phenomenon—the persistent $\approx 8.95\%$ overshoot of truncated Fourier series near a jump discontinuity—is classically derived via the Dirichlet kernel and the sine integral $\mathrm{Si}(\pi)$. This paper does not replace that derivation; it offers a phasor-kinematic rephrasing aimed at engineering intuition.

For the nth Fourier harmonic with amplitude A_n and angular speed $\omega_n = n$, define the *phasor tip speed*

$$v_n := A_n \, \omega_n = n \, A_n.$$

For a piecewise smooth periodic function with a jump of magnitude J, the classical coefficient asymptotic $A_n \sim |J|/(\pi n)$ immediately gives

$$v_n \longrightarrow \frac{|J|}{\pi}.$$

The jump-carrying phasors shrink in radius like $1/n$ while rotating n times faster; their tip speeds converge to a nonzero constant. This provides a direct phasor-language answer to why the Gibbs overshoot height persists even as the overshoot region narrows.

We show that at the discontinuity, the individual velocity contributions of all harmonics are asymptotically aligned—they point in the same direction in the complex plane—producing constructive superposition of the resultant tip velocity. At the Gibbs peak, these equal-length vectors have fanned into a semicircular arrangement, providing a geometric route to the classical overshoot constant $\mathrm{Si}(\pi)/\pi$. We identify a $\pi/2$ phase lag between the velocity maximum and the displacement maximum, analogous to the phase relationship in simple harmonic motion.

We restate the classical Gibbs formula as a product of the tip-speed scale and the universal sinc-accumulation integral $\mathrm{Si}(\pi)$, extend the picture to signals with multiple jumps via interference envelopes, and describe a practical diagnostic: the scaled spectrum $n\,|c_n|$ serves as a quick indicator of jump-type singularities in spectral data.

The core intuition described in this paper arose independently during the author's engineering studies at Bergen University College (2007–2010) through visualisation of Fourier series as systems of rotating phasors, and was formalised in 2026.

1 Introduction

When a truncated Fourier series is evaluated near a jump discontinuity, its partial sums overshoot the midpoint value by approximately 9% of the jump magnitude, independent of how many terms are retained. This is the Gibbs phenomenon, observed by Wilbraham (1848) and widely disseminated by Gibbs (1899) [1]. The classical analysis uses the Dirichlet kernel and yields the universal overshoot fraction

$$\frac{\mathrm{Si}(\pi)}{\pi} - \frac{1}{2} \approx 0.0895, \qquad \mathrm{Si}(x) := \int_0^x \frac{\sin u}{u}\, du.$$

This result is settled [2, 3].

What is less commonly presented is a phasor-kinematic interpretation that explains, in rotating-vector language, *why* the overshoot persists even as its width shrinks to zero. The key observation is classical but rarely named: the product $n\,A_n$ converges to a nonzero constant at a jump. The geometric perspective adopted here—treating the Fourier series as a chain of rotating vectors whose configurations encode the waveform—is in the spirit of Needham's visual approach to complex analysis [4]. That is the content of this paper.

Scope and intent

This paper restates known Fourier asymptotics in phasor language. The mathematical content is not new; the contribution is interpretive. We aim to provide:

- a named quantity ($\mathrm{PTS}_n = n\,A_n$) that captures the scale-invariant feature of jump-carrying harmonics,
- a phasor-kinematic answer to why the Gibbs overshoot height persists,
- a resultant-velocity perspective showing that constructive alignment of harmonic velocities drives the overshoot,
- a geometric interpretation via semicircular vector fans connecting the phasor picture to the classical $\mathrm{Si}(\pi)/\pi$ constant,
- a factorisation viewpoint for the classical Gibbs formula, and
- a practical spectral diagnostic for detecting jump singularities.

2 Phasor Representation of Fourier Series

Let $f\colon [-\pi,\pi] \to \mathbb{R}$ be square-integrable and extended 2π-periodically. Its Fourier series is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n \cos nx + b_n \sin nx\big),$$

with

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx.$$

In the phasor picture, the nth harmonic is a vector of radius

$$A_n = \sqrt{a_n^2 + b_n^2}$$

rotating in the complex plane at angular speed $\omega_n = n$.

Definition 2.1 (Phasor tip speed)**.** The *phasor tip speed* of the nth harmonic is

$$\mathrm{PTS}_n(f) := n\,A_n.$$

This is the linear speed of the tip of the nth rotating phasor—radius times angular speed, the same kinematic relation as $v = r\omega$ for a point on a spinning disc.

3 Jump Asymptotics and Tip-Speed Scaling

3.1 Complex coefficient form

The complex Fourier coefficients are

$$c_n := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

For real-valued f, the relation $A_n = 2|c_n|$ holds for $n \geq 1$.

Proposition 3.1 (Single-jump asymptotics)**.** *Let f be piecewise C^1 as a 2π-periodic function with a single jump discontinuity of magnitude $J = f(x_0^+) - f(x_0^-)$ at $x_0 \in (-\pi,\pi)$. Then as $|n| \to \infty$,*

$$c_n = \frac{J}{2\pi i n}\,e^{-inx_0} + o\!\left(\frac{1}{n}\right).$$

Consequently, for $n \geq 1$,

$$A_n = \frac{|J|}{\pi n} + o\!\left(\frac{1}{n}\right), \qquad \mathrm{PTS}_n(f) = n\,A_n \longrightarrow \frac{|J|}{\pi}.$$

Proof. Integration by parts on smooth subintervals yields the jump contribution as the leading boundary term. Taking moduli and multiplying by n gives the result. □

Theorem 3.2 (Tip-speed scaling at a jump)**.** *Under the hypotheses of theorem 3.1,*

$$\lim_{n\to\infty} \mathrm{PTS}_n(f) = \frac{|J|}{\pi}.$$

Remark 3.3. This is a restatement of the classical asymptotic $A_n \sim |J|/(\pi n)$. The limit $|J|/\pi$ is not a conservation law; it is a compact way to express that jump-carrying harmonics have a nonvanishing tip-speed scale.

Remark 3.4 (Why PTS constancy is not conservation of angular momentum)*.* The visual resemblance between PTS constancy and conservation of angular momentum is striking: in both cases, a shorter arm compensates with faster rotation to maintain a constant product. The distinction is structural.

In a physical system obeying angular momentum conservation (e.g. a spinning ice skater retracting her arms), a single system evolves in time: the moment of inertia I decreases, the angular velocity ω increases, and $L = I\omega$ is conserved. This is a dynamical constraint arising from rotational symmetry via Noether's theorem.

In the phasor chain, each harmonic is an independent basis function with a *fixed* radius A_n and a *fixed* angular speed n. Nothing evolves; nothing is exchanged between harmonics. The nth arm does not "speed up" because the $(n-1)$th arm "slowed down." The equality of tip speeds is a structural fingerprint of the discontinuity: the jump forces $A_n \sim |J|/(\pi n)$, and the product $n \cdot |J|/(\pi n) = |J|/\pi$ is algebraic cancellation, not dynamical compensation.

PTS constancy is an asymptotic scaling property of the Fourier coefficients at a jump—not a conservation law.

3.2 Why the overshoot height persists: the phasor answer

The standard question posed to students is: if each harmonic's amplitude shrinks as $N \to \infty$, why doesn't the overshoot shrink as well?

The phasor answer is direct. Near the jump, the partial sum builds up through a chain of rotating vectors. Each new phasor is smaller, but it rotates faster. The overshoot is governed not by the size of the individual phasors, but by the *speed* at which their tips traverse the complex plane. Since $\mathrm{PTS}_n \to |J|/\pi \neq 0$, each additional phasor contributes a nonvanishing "sweep" near the jump, and the cumulative overshoot does not vanish.

3.3 Resultant velocity and constructive alignment

The per-harmonic tip speed $\mathrm{PTS}_n = nA_n$ describes each arm in isolation. What drives the overshoot is the *resultant* velocity of the chain tip—the vector sum of all individual

tangential velocities.

The velocity contribution of the nth harmonic to the partial sum is

$$\frac{d}{dx}\left(c_n\, e^{inx}\right) = in\, c_n\, e^{inx},$$

and the total velocity of the Nth partial sum at position x is

$$S_N'(x) = \sum_{n=-N}^{N} in\, c_n\, e^{inx}.$$

At the discontinuity $x = x_0$, the leading-order terms satisfy $c_n \sim J/(2\pi in)\, e^{-inx_0}$, so

$$in\, c_n\, e^{inx_0} \;\sim\; in \cdot \frac{J}{2\pi in} \cdot e^{-inx_0} \cdot e^{inx_0} \;=\; \frac{J}{2\pi}.$$

Each harmonic's velocity contribution is asymptotically *real and positive* at the jump—the velocity vectors are aligned. The resultant velocity is therefore approximately $N \cdot J/(2\pi)$, which grows with N. This constructive alignment is why the partial sum traverses the overshoot region at increasing speed, producing a sharper but equally tall spike.

Away from the jump, the phase factors $e^{in(x-x_0)}$ rotate the individual velocity contributions, destroying the alignment. The vectors scatter and partially cancel. This is the kinematic mechanism behind the rapid decay of the Gibbs ringing away from the discontinuity.

3.4 The $\pi/2$ phase lag: velocity peak versus displacement peak

A subtlety visible in the phasor chain is that the maximum resultant velocity does not coincide with the Gibbs overshoot peak. The velocity is maximised at $x = x_0$ (the discontinuity itself), while the overshoot peak occurs at $x \approx x_0 + \pi/N$—*after* the velocity maximum.

This is the standard phase relationship between velocity and displacement in any oscillating system. The partial sum $S_N(x)$ near the jump can be written in terms of the Dirichlet kernel D_N:

$$S_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\, D_N(x-t)\, dt.$$

Near x_0, the dominant contribution comes from the jump, and the overshoot height is governed by the integral

$$\int_0^{\pi/N} D_N(u)\, du \;\approx\; \int_0^{\pi} \frac{\sin v}{v}\, dv \;=\; \mathrm{Si}(\pi),$$

after the substitution $v = Nu$. The kernel $D_N(u)$ plays the role of a velocity profile: it peaks at $u = 0$ and its integral peaks where the kernel first crosses zero, at $u \approx \pi/N$.

This is the same $\pi/2$ phase relationship (in the rescaled variable $v = Nu$) that governs position and velocity in simple harmonic motion: velocity peaks at equilibrium, displacement peaks a quarter-cycle later. The Gibbs overshoot occurs not where the phasor velocities are maximally aligned, but where their accumulated displacement is greatest—a quarter-cycle downstream.

3.5 The semicircular vector fan

At the Gibbs peak ($x \approx x_0 + \pi/N$), the individual velocity vectors form a characteristic geometric pattern that yields a purely visual derivation of the overshoot constant.

Observation 3.5 (Semicircular fan at the Gibbs peak). Consider the N-term partial sum of a function with a single jump J at x_0. At the overshoot peak $x = x_0 + \pi/N$, the velocity contribution of the nth jump-carrying harmonic has phase

$$\phi_n = n \cdot \frac{\pi}{N}$$

relative to the alignment at x_0. For harmonics $n = 1, \ldots, N$, the total phase spread is

$$\phi_N - \phi_1 = (N-1)\frac{\pi}{N} \;\to\; \pi \qquad \text{as } N \to \infty.$$

Thus the N velocity vectors, each of approximately equal magnitude $|J|/(2\pi)$, are arranged in a fan spanning a semicircle.

The overshoot has a geometric interpretation in terms of this fan:

- The **arc length** of the semicircle is the scalar sum of all velocity magnitudes: $N \cdot |J|/(2\pi) \approx N|J|/(2\pi)$. If the vectors were still aligned, this would be the displacement per unit x.
- The **chord** of the semicircular fan is the magnitude of the resultant (vector sum). For N equal vectors spanning an angle π, the chord-to-arc ratio is

$$\frac{\text{chord}}{\text{arc}} = \frac{|\sum \mathbf{v}_n|}{\sum |\mathbf{v}_n|} \;\to\; \frac{2}{\pi} \qquad \text{as } N \to \infty.$$

The overshoot level above $f(x_0^-)$ is determined by the *integrated* displacement from x_0 to the peak, which involves the accumulation of these fanning vectors. The precise integral yields the sine integral $\mathrm{Si}(\pi)$, recovering the classical Gibbs constant.

The semicircular fan thus provides a geometric picture: the Gibbs overshoot arises because N equal-length vectors fan from full alignment (at the jump) into a semicircle (at the peak). The accumulated displacement during this fanning process exceeds the target value by exactly the factor $\mathrm{Si}(\pi)/\pi \approx 1.0895$.

3.6 Vector polygon sequence and the waveform

As the phase variable x advances beyond the discontinuity, the equal-length velocity vectors do not stop at a semicircle. They continue to fan, passing through a sequence of recognisable geometric configurations.

For a partial sum with N harmonics, the phases $\phi_n = n(x - x_0)$ spread further as x increases. The velocity vector polygon evolves through:

1. **Full alignment** at $x = x_0$: all vectors point in the same direction. Maximum resultant velocity.
2. **Semicircular fan** at $x \approx x_0 + \pi/N$: the Gibbs peak. Vectors span π radians.
3. **Symmetric polygon phases**: as x increases further, the vectors form progressively more closed figures—for $N = 5$ harmonics (at $k = 1, 3, 5, 7, 9$), one observes pentagons, triangles, and star (pentagram) configurations.
4. **Maximum cancellation** at $x = \pi/2$: the phases are distributed so that vectors nearly cancel in pairs. This corresponds to the flat settled portion of the waveform.
5. **Rebuilding**: as x approaches the next jump, the vectors re-align in the opposite direction.

The entire waveform is thus encoded in the geometry of the velocity vector polygon. High-amplitude regions correspond to open, aligned polygons; low-amplitude settled regions correspond to closed or cancelling polygons. The Gibbs overshoot is the brief interval during which the polygon transitions from full alignment to its first symmetric closure—the semicircle.

Remark 3.6 (The waveform as vector choreography). This polygon perspective converts the Fourier partial sum from a numerical summation into a geometric narrative: the waveform is the shadow of a sequence of opening and closing vector figures. While this observation is implicit in any phasor-chain visualisation, naming the specific polygon stages (alignment $\to$ semicircle $\to$ symmetric closure $\to$ cancellation $\to$ re-alignment) provides a structural vocabulary for describing *where* and *why* each feature of the waveform arises.

4 Gibbs Overshoot in Prefactor–Kernel Form

Theorem 4.1 (Gibbs overshoot)**.** *Under the hypotheses of theorem 3.1, the overshoot of the Nth partial sum above the midpoint value $f(x_0^-) + J/2$ satisfies*

$$\Delta = J\left(\frac{\mathrm{Si}(\pi)}{\pi} - \frac{1}{2}\right) \approx 0.0895\, J$$

in the $N \to \infty$ Gibbs limit.

Remark 4.2 (Prefactor–kernel factorisation). In the classical derivation, the jump enters the overshoot formula as a prefactor J/π multiplied by the universal sinc-accumulation integral $\mathrm{Si}(\pi)$. By theorem 3.2, the magnitude $|J|/\pi$ is exactly the asymptotic tip-speed scale of the jump-carrying harmonics. Thus the classical Gibbs formula can be read as

$$\text{overshoot level above } f(x_0^-) \;=\; \underbrace{\frac{|J|}{\pi}}_{\text{tip-speed scale}} \;\times\; \underbrace{\mathrm{Si}(\pi)}_{\text{sinc accumulation}} \quad .$$

This factorisation is notational, not a new derivation. Its value is that it identifies the jump prefactor with a named phasor-kinematic quantity.

Corollary 4.3 (Universal fractional overshoot)**.** *The fractional overshoot* $\Delta/J = \mathrm{Si}(\pi)/\pi - 1/2 \approx 0.0895$ *is independent of* J*, of the jump location, and of the smooth behaviour of* f *away from the jump.*

5 Multiple Jumps and Interference Envelopes

For functions with K jumps of magnitudes $J_1, \ldots, J_K$ at locations $x_1, \ldots, x_K$, the large-n complex coefficients are

$$c_n \sim \frac{1}{2\pi i n} \sum_{k=1}^{K} J_k \, e^{-inx_k}.$$

The real-basis tip speed is therefore

$$\mathrm{PTS}_n = n\, A_n \;\sim\; \frac{1}{\pi} \left| \sum_{k=1}^{K} J_k \, e^{-inx_k} \right| .$$

Unlike the single-jump case, PTS_n need not converge to a constant. Instead it traces an *interference envelope* determined by the relative phases e^{-inx_k} of the jump contributions. Constructive interference produces tip-speed maxima; destructive interference produces near-zero values.

This is the familiar superposition principle applied to the jump contributions in the spectral domain. For the square wave (two equal jumps, symmetrically placed), all odd-n phase factors align constructively, giving $\mathrm{PTS}_n = 4/\pi$ for odd n and $\mathrm{PTS}_n = 0$ for even n.

6 Tip-Speed Spectrum as a Jump Diagnostic

For smooth functions, Fourier coefficients decay faster than $1/n$ (e.g. $1/n^2$ for a continuous function with a corner, exponentially for analytic functions). Therefore the scaled spectrum $n\,|c_n|$ decays to zero for smooth signals. At a jump, it plateaus.

This gives a practical diagnostic:

Behaviour of $n\,|c_n|$ as $n \to \infty$	**Signal regularity**
$n\,|c_n| \to 0$	smoother than a jump (at least continuous)
$n\,|c_n| \to$ nonzero constant	single jump present
$n\,|c_n|$ oscillates with nonzero envelope	multiple jumps with phase interference

Remark 6.1 (Practical usage). In a discrete setting (DFT of N samples), one computes $k\,|X_k|/N$ for bin indices k and checks whether this quantity plateaus or decays in the upper portion of the spectrum. A plateau indicates that the signal contains step-like discontinuities or very sharp transients. This is a first-pass diagnostic, not a replacement for detailed regularity analysis, but it is cheap and informative.

7 Illustrative Example: The Square Wave

Consider the square wave

$$f(x) = \operatorname{sgn}(\sin x), \qquad x \in [-\pi, \pi],$$

extended periodically. Its nonzero Fourier coefficients are $b_n = 4/(n\pi)$ for odd n, giving

$$A_n = \frac{4}{n\pi}, \qquad \mathrm{PTS}_n = \frac{4}{\pi} \approx 1.273 \qquad (n \text{ odd}).$$

Reconciliation with the single-jump formula. Each individual jump has magnitude $J = 2$, so theorem 3.2 predicts $\mathrm{PTS}_n \to 2/\pi \approx 0.637$ per jump. The square wave has two jumps per period at $x = 0$ and $x = \pi$. For odd n, the phase factors $e^{-in\cdot 0} = 1$ and $e^{-in\pi} = (-1)^n = -1$ combine with equal jump magnitudes $+2$ and -2 to give a coherent sum of magnitude 4. Thus $\mathrm{PTS}_n = 4/\pi$ is the constructive-interference case of Section 5, not a contradiction.

Gibbs overshoot. The overshoot at each jump remains

$$\Delta = 2\left(\frac{\mathrm{Si}(\pi)}{\pi} - \frac{1}{2}\right) \approx 0.179,$$

which is 8.95% of the jump magnitude $J = 2$.

8 Conclusion

We defined the phasor tip speed $\mathrm{PTS}_n = n\,A_n$ and restated the classical jump asymptotic $A_n \sim |J|/(\pi n)$ as the scaling limit $\mathrm{PTS}_n \to |J|/\pi$. This provides a phasor-kinematic

answer to why the Gibbs overshoot height persists: the jump-carrying harmonics have radii decaying like $1/n$ and angular speeds growing like n, producing a nonvanishing tip-speed scale that sustains the overshoot even as individual amplitudes shrink.

The resultant velocity of the phasor chain—the vector sum of all harmonic velocity contributions—reveals the mechanism more precisely. At the discontinuity, all velocity contributions are asymptotically aligned: every harmonic pushes the chain tip in the same direction. The tip accelerates through the target value. As the phase advances, the velocity vectors fan out, and at the Gibbs peak ($x \approx x_0 + \pi/N$), they span a semicircle. The overshoot is the accumulated displacement during this fanning process. The displacement peak lags the velocity peak by a quarter-cycle in the rescaled variable—the same $\pi/2$ phase relationship that governs position and velocity in harmonic motion.

The semicircular vector fan provides a geometric route to the classical Gibbs constant: N equal-length vectors spanning π radians yield an accumulated displacement proportional to $\mathrm{Si}(\pi)$, recovering the 8.95% overshoot without direct appeal to the Dirichlet kernel. As the phase continues to advance, the vector polygon passes through symmetric closure configurations (pentagons, triangles, stars for small N) before reaching maximum cancellation at the settled waveform value. The entire waveform is encoded in the geometry of these evolving vector figures.

PTS constancy is not a conservation law. It is an asymptotic scaling property—a structural fingerprint of the discontinuity, not a dynamical balance. Each harmonic's radius and angular speed are fixed from the outset; nothing is exchanged between harmonics.

The classical Gibbs overshoot formula admits a factorisation into the tip-speed scale and the universal sinc-accumulation integral $\mathrm{Si}(\pi)$. For signals with multiple jumps, the tip-speed spectrum traces an interference envelope governed by the jump phases. The scaled spectrum $n\,|c_n|$ serves as a practical diagnostic for the presence of jump-type singularities.

All results are restatements of standard Fourier asymptotics. The value, if any, is in the phasor language, the geometric perspective, and the diagnostic viewpoint.

References

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